



1. Project Setup

In your terminal:

```
composer create-project laravel/laravel khqr-bakong
cd khqr-bakong
composer require khqr-gateway/bakong-khqr-php
composer require simplesoftwareio/simple-qrcode
```

Then configure `.env` with your **app name**, **database**, **BAKONG_TOKEN**, etc.

2. Database Setup

Run migrations and create a simple `products` table.

```
php artisan make:model Product -m
```

Then open the migration file (`database/migrations/xxxx_create_products_table.php`) and add:

```
public function up(): void
{
    Schema::create('products', function (Blueprint $table) {
        $table->id();
        $table->string('name');
        $table->text('description')->nullable();
        $table->decimal('price', 10, 2);
        $table->string('image')->nullable();
        $table->timestamps();
    });
}
```

Run:

```
php artisan migrate
```

Then seed some sample products:

```
Product::create([
```

```

        'name' => 'iPhone 17 Pro',
        'description' => '128GB, Blue Titanium',
        'price' => 200,
        'image' => 'https://i.ebayimg.com/images/g/EDEAAeSwLb9owuv8/s-l1600.webp'
    ]);
    Product::create([
        'name' => 'AirPods Pro 3',
        'description' => 'Wireless earphones with noise cancelling',
        'price' => 150,
        'image' => 'https://sm.mashable.com/t/mashable_sea/article/a/apple-airp/apple-airpods-pro-3-every-single-thing-we-know-so-far_uj6k.1248.jpg'
    ]);

```

php artisan db:seed

3. ProductController

Create controller:

```
php artisan make:controller ProductController
```

Add code:

```

<?php

namespace App\Http\Controllers;

use App\Models\Product;

class ProductController extends Controller
{
    public function index()
    {
        $products = Product::all();
        return view('products.index', compact('products'));
    }

    public function show($id)
    {
        $product = Product::findOrFail($id);
        return view('products.show', compact('product'));
    }
}

```



4. PaymentController (Bakong Integration)

php artisan make:controller PaymentController

Add code:

```
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use App\Models\Product;
use KHQR\BakongKHQR;
use KHQR\Helpers\KHQRData;
use KHQR\Models\IndividualInfo;
use SebastianBergmann\Environment\Console;

class PaymentController extends Controller
{
    public function checkout($id)
    {
        $product = Product::findOrFail($id);

        // Your merchant info (replace with real Bakong account)
        $merchant = new IndividualInfo(
            bakongAccountID: 'vannak_dim@cadi',
            merchantName: 'VANNAK DIM',
            merchantCity: 'Phnom Penh',
            currency: KHQRData::CURRENCY_KHR,
            amount: $product->price
        );

        $qrResponse = BakongKHQR::generateIndividual($merchant);

        return view('products.checkout', [
            'product' => $product,
            'qr' => $qrResponse->data['qr'] ?? null,
            'md5' => $qrResponse->data['md5'] ?? null,
        ]);
    }

    public function verifyForm()
    {
        return view('payments.verify');
    }

    public function verifyTransaction(Request $request)
```

```

{
    $request->validate([
        'md5' => 'required|string',
    ]);

    try {
        // Your Bakong API token from https://api-bakong.nbc.gov.kh/register
        $token = env('BAKONG_TOKEN'); // put it in .env
        $bakong = new \KHQR\BakongKHQR($token);

        $result = $bakong->checkTransactionByMD5($request->md5);

        return response()->json($result);
        // return view('payments.result', ['result' => $result]);
    } catch (\Exception $e) {
        // return back()->withErrors(['error' => $e->getMessage()]);
        return response()->json(['error' => $e->getMessage()], 500);
    }
}

public function paymentResult()
{
    return view('payments.result');
}
}

```



5. Routes

Edit routes/web.php:

```

<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProductController;
use App\Http\Controllers\PaymentController;

Route::get('/', [ProductController::class, 'index'])->name('home');
Route::get('/product/{id}', [ProductController::class, 'show'])->name('product.show');
Route::post('/checkout/{id}', [PaymentController::class, 'checkout'])->name('checkout');
Route::get('/verify', [PaymentController::class, 'verifyForm'])->name('verify.form');
Route::post('/verify', [PaymentController::class, 'verifyTransaction'])->name('verify.transaction');
Route::get('/payments/result', [PaymentController::class, 'verifyTransaction'])->name('payments.result');

```



6. Views

`resources/views/products/index.blade.php`

```
@extends('layouts.app')

@section('content')
<div class="container">
    <h1>Product List</h1>
    <div class="row">
        @foreach($products as $product)
            <div class="col-md-4">
                <div class="card mb-3">
                    name }}">
                    <div class="card-body">
                        <h5>{{ $product->name }}</h5>
                        <p>{{ $product->description }}</p>
                        <p><strong>{{ number_format($product->price, 0) }} </strong></p>
                        <a href="{{ route('product.show', $product->id) }}" class="btn
btn-primary">Buy</a>
                    </div>
                </div>
            </div>
        @endforeach
    </div>
</div>
@endsection
```

`resources/views/products/show.blade.php`

```
@extends('layouts.app')

@section('content')
<div class="container">
    <h1>{{ $product->name }}</h1>
    
    <p>{{ $product->description }}</p>
    <p><strong>{{ number_format($product->price, 0) }} </strong></p>

    <form action="{{ route('checkout', $product->id) }}" method="POST">
        @csrf
        <button class="btn btn-success">Generate KHQR to Pay</button>
    </form>
</div>
@endsection
```

```
        </form>
    </div>
@endsection
```

resources/views/products/checkout.blade.php

```
@extends('layouts.app')

@section('content')
    <div class="container text-center">
        <h1>Scan KHQR to Pay</h1>
        <p><strong>{{ $product->name }}</strong> - {{ number_format($product->price,
2) }} $</p>

        @if ($qr)
            {!! QRCode::size(300)->generate($qr) !!}
            <p class="mt-3">MD5: {{ $md5 }}</p>
            <p class="text-muted">Scan this QR code using Bakong App to make a
payment.</p>
        @else
            <p class="alert alert-danger">⚠ Failed to generate KHQR.</p>
        @endif

        {{-- Countdown Timer --}}
        <div class="mt-4">
            <h3 id="countdown" class="text-danger fw-bold">120</h3>
            <p class="text-muted">This page will expire in <span
id="seconds">120</span> seconds.</p>
        </div>

        <a href="{{ route('home') }}" class="btn btn-secondary mt-4">Back to Shop</a>
    </div>

    {{-- Countdown Script --}}
    <script>
        let timeLeft = 120; // seconds
        const countdownElement = document.getElementById('countdown');
        const secondsText = document.getElementById('seconds');

        const timer = setInterval(() => {
            timeLeft--;
            countdownElement.textContent = timeLeft;
            secondsText.textContent = timeLeft;
            if (timeLeft > 0) {
                fetch("{{ route('verify.transaction') }}", {
                    method: "POST",
```

```

        headers: {
            "Content-Type": "application/json",
            "X-CSRF-TOKEN": "{{ csrf_token() }}"
        },
        body: JSON.stringify({
            md5: "{{ $md5 }}"
        })
    })
    .then(response => response.json())
    .then(data => {
        console.log(data);
        if (data.responseCode === 0) {
            clearInterval(timer);
            alert("Transaction successful!");
            window.location.href = "{{ route('home') }}";
        } else if (data.failed) {
            alert("Transaction failed. Please try again.");
        }
    })
    .catch(error => console.error('Error:', error));
}
if (timeLeft <= 0) {
    clearInterval(timer);
    window.location.href = "{{ route('home') }}";
}
}, 1000);
</script>
@endsection

```

resources/views/layouts/app.blade.php

```

<!DOCTYPE html>
<html>
<head>
    <title>Bakong KHQR Shop</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
    @yield('content')
</body>
</html>

```

7. Run the Project

```
php artisan serve
```

Visit:

👉 <http://127.0.0.1:8000>

You'll see product cards — click **Buy**, then it will generate and show a **KHQR code** using your Bakong account ID.

Notes

- Replace 'yourbakongid@bankcode' with your actual **Bakong Merchant Account ID**.
- If you want to **verify payments**, you can extend the `PaymentController` using:
 - `$bakong = new BakongKHQR('your_api_token_here');`
 - `$bakong->checkTransactionByMD5($md5);`
- To test with real Bakong QR scanning, use the **Bakong app** and scan the generated QR image.